I. Choice (3 points X 15)

1. Information is transmitted ***between*** neurons by:
A. vesicular release, diffusion, and reception of molecules called neurotransmitters.
B. ion flow across gap junctions.
C. action potentials.
D. A and B
E. All of the above

2. Why do action potentials usually travel unidirectionally down an axon?
A. Delayed activation of K^+ channels
B. Inactivation of Na^+ channels
C. Myelin prevents travel in the opposite direction.
D. Action potentials are all-or-none.

3. What is the **input** nucleus of the basal ganglia?

A. Striatum
B. VTA
C. Substantia nigra pars reticulata
D. Subthalamic nucleus
E. Globus pallidus

4. Addiction is mostly related to:

A. Glutamate
B. GABA
C. Epinephrine
D. Dopamine

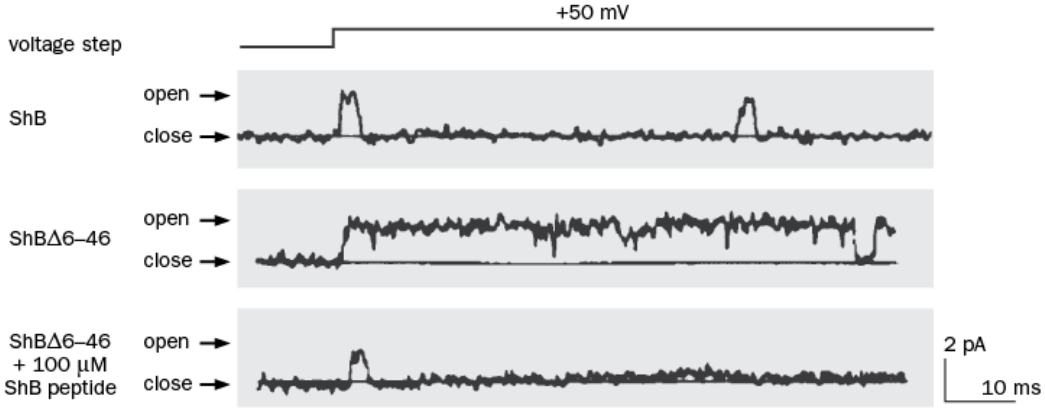
5. How is LTP generated? Choose the **wrong** answer.

A. High-frequency stimulation
B. Low-frequency stimulation
C. Increase in intracellular Ca^{2+}
D. Activation of CaMKII

6. Which ion is directly related to vesicle release?
A. Na+
B. K+
C. Ca2+
D. Cl-

7. **Figure** shows a patch clamp recording from the Shaker channel (ShB). What is the presumed

molecular mechanism for inactivation of the channel?



A. Several positively charged amino acids in S4
B. Several negatively charge amino acids in S4
C. The N-terminal 'ball-and-chain'
D. A block by Na+ ions

8. What is the 'blind spot'? Select all that are correct.

A. The place in the retina in which all axons exit the retina to go to the brain.
B. The place the lens focuses light on the retina.
C. The location in the retina where light has a direct path to the photoreceptors.
D. The location in the retina with the highest density of capillaries.

9. Rhodopsin is what kind of protein?

A. Voltage-dependent ion channel
B. Ion co-transporter
C. Na+ pump
D. G-protein-coupled receptor

10. Hubel and Wiesel identified simple cells and complex cells in the visual cortex. Simple cells respond to bars of light in a particular orientation. For example, stimulation by a bar of light in the center of the visual field could excite the V1 neuron but moving the bar of light to either side of the center will inhibit the neuron. What anatomical structure did Hubel and Wiesel propose that would confer this response on the V1 neuron?

A. The receptive field of the simple cell is the result of input from many complex cells within the visual cortex.
B. The receptive field of the simple cell is the integration of inputs from many LGN neurons that have OFF-center/ON-surround responses.
C. The receptive field of the simple cell is the integration of inputs from many LGN neurons that have ON-center/OFF-surround responses.
D. The receptive field of the simple cell reflects complex processing in cortical microcircuits.

11. Olfactory sensory neurons (ORNs) expressing the same odorant receptor:
A. are spatially clustered in the nasal epithelium.

B. are distributed randomly within one of four zones in the nasal epithelium.
C. are distributed randomly throughout the entire nasal epithelium.
D. are distributed randomly throughout the olfactory bulb.
E. None of the above

12. Which of the following statements is not true of taste in mammals?

A. Each sensory neuron expresses multiple types of receptors.
B. Sensory neuron activity reflects the biological significance of the odorant.
C. A large number of chemical compounds can be discriminated from each other.
D. All of the above
E. None of the above

13. The amnesic patient, H.M., could still learn motor tasks, like learning to draw while looking in a mirror. What kind of memory is this? Choose all that apply.

A. Explicit memory
B. Implicit memory
C. Habituation
D. Sensitization

14. What do you predict would happen if the activity of hypocretin neurons in the hypothalamus were genetically silenced?

A. The animal would spend more time asleep.
B. The animal would spend more time awake.
C. The animal would have an increased probability of making a transition from REM sleep to wake.
D. The animal would not be able to sleep at all.

15. What are two molecules involved in short-term signaling from the gastrointestinal system?

A. Ghrelin and insulin
B. Insulin and leptin
C. Insulin and cholecystokinin
D. Ghrelin and cholecystokinin
E. Dopamine and ghrelin
F. Dopamine and insulin

II. Short Answer (5 questions, 25 points in total)

1. How does high-frequency stimulation induce LTP (long-term potentiation)?

2. If myelin makes action conduction velocity faster, why are axons not completely covered with myelin? That is, what would happen to the electrical signal if the axon was completely covered in myelin and had no Nodes of Ranvier?

3. In the photoreceptor, stimulation of light results in an increase or decrease in glutamate release? Explain your answer.

4. After your exam, you go home and lay on the bed and eat some snacks. Which division of the autonomic nervous system is most active and what would happen to your heart rate?

5. There are many neurodegenerative diseases. List one and describe its main symptoms.

III. Essay questions (5 questions, 30 points in total)

1. Describe, in general terms, experiments that show that Ca2+ is both necessary and sufficient for neurotransmitter release. Start with the assumption that you are recording from the squid giant synapse in which you can stimulate the presynaptic neuron and record the response in the postsynaptic neuron. State which experiment shows the necessity and which shows sufficiency.

2. Visual prosthesis is a new technology that can restore some blind vision. There is a lot of research on the visual prosthesis to help people recover some vision. Describe a target and how you could think about how the prosthesis would work. The details are not as important as the rationale for your idea. Include some of your assumptions.